



Fig. 13: Single-stage and multi-stage TE modules

The heat transfer according to the Peltier effect takes place in the direction of flow of charge carriers.

A TE module literally pumps heat from one place to another, heating one face and cooling the other in the process. Whether a particular face becomes hot or cold depends upon the direction of current flow through the TE module. Thus, by controlling the magnitude and direction of the current drive to the TE module, it can be used as a control element in a temperature stabilisation circuit. The TE module pumps the heat dissipated in the laser diode on to a heat sink mounted on the supposedly hot face of the TE module. The heat is transferred from heat sink to the ambient.

TE modules are characterised by maximal performance parameters, namely, maxi-

schematic of TE-module-based temperature control circuit. Key building blocks of the drive and control circuit include temperature sensor, error amplifier, error signal processor and a bipolar output driver circuit. Bipolar output driver provides the required power to drive the TE module. The output stage in most cases is designed to allow flow of drive current in either direction through the TE module. This enables both cooling and heating of the device to maintain its temperature at the specified value, regardless of ambient temperature being higher or lower.

Negative temperature coefficient (NTC) thermistor is the most commonly used temperature sensor due to its high sensitivity. RTD and semiconductor sensors have better linearity but suffer from poor sensitivity. Semiconductor sensors are also available in integrated circuit form producing a current linearly related to absolute temperature. AD 590 is one such sensor. Balanced bridge circuit configuration that generates a differential output is generally used. Error amplifier is configured around an op-amp. Error-signal processor could be anything from simple on-off controller to a proportional controller or proportional-integral (PI) or even fully digital PI-differential (PID) controller.

On-off controller is never used in practice. In proportional control, the drive signal to the TE module is proportional to the difference between the actual and desired temperatures. In a proportional controller, there is always a residual error even after the controller has settled to the final state. This error is proportional to the difference between the desired and the actual temperature and is inversely proportional to the gain of the control loop.

The problem of residual error of a proportional controller can be overcome by the addition of an integrator in the control loop. The result is a PI controller. One disadvantage of PI control is that it would be slow to respond

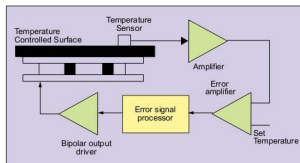


Fig. 14: Schematic arrangement of TE temperature-control circuit

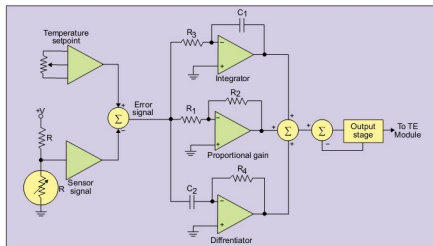


Fig. 15: PID controller and other building blocks

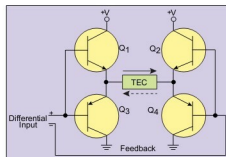


Fig. 16: Output stage configured around half-bridge circuit

imum temperature difference ΔT_{MAX} at zero heat load, maximum heating capacity Q_{MAX} for zero temperature difference, device current I_{MAX} measured at ΔT_{MAX} , terminal voltage V_{MAX} corresponding to I_{MAX} with no heat load and coefficient of performance (COP), which is the ratio of pumped heat load to electrical power supplied to the device.

Fig. 14 shows the basic block